Introduction to





with Application to Bioinformatics

- Day 4

Quiz: Review Day 3

Go to Canvas, `Modules -> Day 4 -> Review Day 3`

~20 minutes

1. What will this code output?

In [ ]:

```
row = 'sofa|2000|buy|Uppsala'
fields = row.split('|')
price = fields[1]
if price == 2000:
    print('The price is a number!')
if price == '2000':
    print('The price is a string!')
```

1. What will this code output?

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row = 'sofa|2000|buy|Uppsala'
fields = row.split('|')
price = fields[1]
if price == 2000:
    print('The price is a number!')
if price == '2000':
    print('The price is a string!')
```

- `split()` always gives a list of **strings**, so `price` is `'2000'`, not `2000`
- A string is never equal to a number: `'2000' == 2000` is `False`
- Convert it if you need a number: `int(fields[1])`

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In [ ]:
```
print(sorted([2000, 30, 100]))
```

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Numbers are sorted by their **value**: from the smallest to the largest.

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print(sorted(['2000', '30', '100']))
```

- Strings are sorted **alphabetically**, one character at a time, like words in a dictionary
- The first characters are `'1'`, `'2'` and `'3'`, so the order is `'100'`, `'2000'`, `'30'`
- Same values, different type, different result!

4. How do you convert the string `'2000'` to a number?

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- `int('2000')` or `float('2000')`

In [ ]:
```
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```

`int` gives a whole number, `float` gives a decimal number.

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In [ ]:
```
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```

`'0.5'` has decimals, so only `float` works: `int('0.5')` gives a `ValueError`.

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`1e9` is scientific notation: $1 \times 10^9 = 1\,000\,000\,000$. `int()` does not understand it:

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In [ ]:
```
int('1e9')
```

7. What value will these conversions give you?

In [ ]:
```
values = [1, 0, '0', '', {}, [], [0]]
for x in values:
    print(repr(x), bool(x))
```

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```

- `0`, the empty string `''` and empty containers (`[]`, `{}`) are `False`
- Everything else is `True`, even the string `'0'`: it is not empty!

8. Container types: which type matches each statement?

- **list**: when order is important
- **dict**: to keep track of the relation between keys and values
- **set**: to check for membership. No order, no duplicates.

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genre_list = ["comedy", "drama", "drama", "sci-fi"]
genre_list
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In [ ]:
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genres = set(genre_list)
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genre_list
```

In [ ]:
```
genres = set(genre_list)
genres
```

In [ ]:
```
genre_counts = {"comedy": 1, "drama": 2, "sci-fi": 1}
genre_counts
```

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genre_list
```

In [ ]:
```
genres = set(genre_list)
genres
```

In [ ]:
```
genre_counts = {"comedy": 1, "drama": 2, "sci-fi": 1}
genre_counts
```

The list keeps the order and the duplicates, the set removes the duplicate `'drama'`, and the dictionary links each genre to its count.

9. Match the following statements with their Python syntax

- `def` : starts a new function definition
- `return` : ends a function
- `print()` : shows the current state of a variable
- `#` : marks that a line should not be executed

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In [ ]:

```
def greet(message):  # def: starts a new function definition
    # print('message is:', message)
    if message == 'Hi':
        print('Hi!')
        return  # return: the function ends here
    print('This was not a Hi')

greet('Hi')
greet('Hello')
```

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- `return` : ends a function
- `print()` : shows the current state of a variable
- `#` : marks that a line should not be executed

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    # print('message is:', message)
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        print('Hi!')
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greet('Hi')
greet('Hello')
```

- With `'Hi'`, `return` ends the function, so the last `print` does not run
- The line starting with `#` is not executed: remove the `#` to see the value of `message`

10. Which string method is used at line 2 to modify `x`?

- `.join()`

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In [ ]:
```
x = ['h', 'e', 'l', 'l', 'o']
y = '_'.join(x)
print(y)
```

10. Which string method is used at line 2 to modify `x`?

- `.join()`

In [ ]:
```
x = ['h', 'e', 'l', 'l', 'o']
y = '_'.join(x)
print(y)
```

- `.join()` is a **string** method: the string `'_'` is put between the elements of the list
- `.split()` does the opposite, from a string back to a list:

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- `.split()` does the opposite, from a string back to a list:

In [ ]:
```
y.split('_')
```

TODAY

Session 1

- More on functions:
    - return statement
    - positional arguments and keyword arguments
    - default and optional arguments

Session 2

- Comments and documentation
- Basic error handling
- User input in python

Session 3

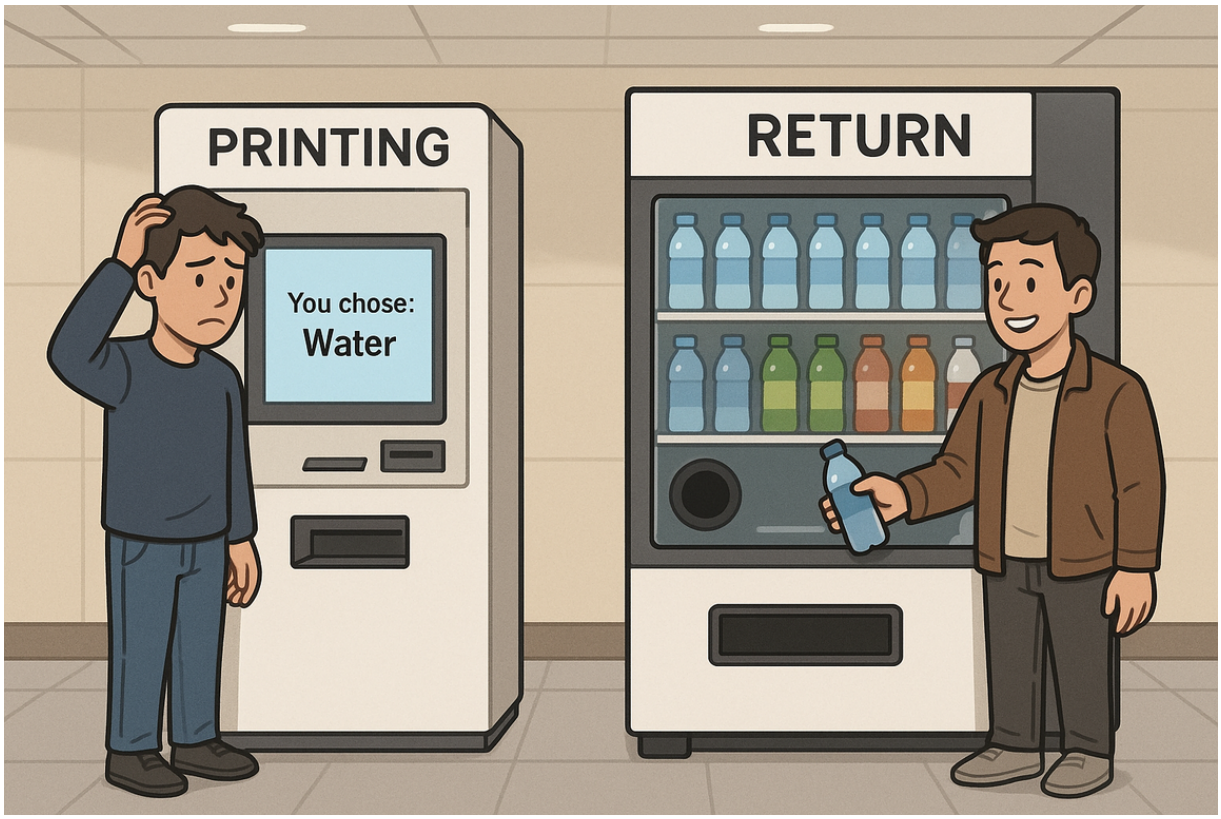
- Pandas - explore your data!

More on functions: `return` statement

`return` vs `print`

Functions that `return` are easier to repurpose than those that `print` their result.

Example



- **Print vending machine:** Only see an information. Cannot use the water
- **Return vending machine:** Have something real in your hands and can use it:
    - Drink it yourself (use the value)
    - Put it in your bag for later (store in a variable)
    - Give it to a friend (pass to another function)

Function with `return` statement

A function that counts the number of occurrences of `'C'` in the argument string.

In [ ]:
```
def cytosine_count(nucleotides):
    count = 0
    for x in nucleotides:
        if x == 'c' or x == 'C':
            count += 1
    return count

count1 = cytosine_count('CATACTAC')
count2 = cytosine_count('tcgcag')
```

- Print the result

In [ ]:
```
print(f"{count1}\n{count2}")
```

- Repurpose the result: Sum the `'C'` occurrences

In [ ]:
```
total_count = count1 + count2
print(f"Total number of C: {total_count}")
```

Functions can return multiple values

In [ ]:

```
# Modify the function below for returning multiple values
def cytosine_count(nucleotides):
    count = 0
    for x in nucleotides:
        if x == 'c' or x == 'C':
            count += 1
    return count
```

Functions without any `return` statement returns `None`

In [ ]:
```
def foo():
    do_nothing = 1

result = foo()
print(f'Return value of foo() = {result}')
```

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Return statement outcome

- Use `return` for all values that you might want to use later in your program or in simple words: **always**

Small detour: Python's value for missing values: `None`

- Default value for optional arguments
- Implicit return value of functions without a `return` statement
- `None` is `None`, not anything else (special value).

In [ ]:
```
None == 0
```

In [ ]:
```
None == False
```

In [ ]:
```
None == ''
```

In [ ]:
```
type(None)
```

More on functions: Keyword arguments

There are two types of arguments:

- **Positional:** values are passed in the same order as parameters are defined in the function
- **Keyword:** values are passed by parameters names while calling a function

Examples

In [ ]:

```
def myFavoriteMovie(pet_number, pet):
    print(f'I have {pet_number} {pet}')

# Positional arguments
myFavoriteMovie(2, "dogs")

# Keyword arguments
myFavoriteMovie(pet_number=3, pet="cats")
```

Why do we use keyword arguments?

- It increases the clarity and readability
- Independent on the order of the values
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In [ ]:
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In [ ]:
```
record.split(maxsplit=2, sep=' ')
```

Both positional and keyword arguments can be used (if it is allowed)

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record = 'gene_id INSR "insulin receptor"'

record.split(' ', maxsplit=2)
```

But there are some exceptions

In [ ]:
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record = 'gene_id INSR "insulin receptor"'

record.split(sep=' ', 2)
```

- Positional arguments must be **in front** of keyword arguments

Restrictions by purpose

General example

In [ ]:
```
def describe_gene(name, sequence, length):
    print(f"Gene: {name}, Sequence: {sequence}, Length: {length}")

describe_gene('BRCA1', 'ATCG', 4)                                  # all pos
describe_gene(name='BRCA1', sequence='ATCG', length=4)             # all key
describe_gene('BRCA1', sequence='ATCG', length=4)                  # mixed
```

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Built-in function example

In [ ]:
```
sorted([1, 4, 100, 5, 6], reverse=True)
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In [ ]:
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```

In [ ]:
```
sorted(iterable, /, *, key=None, reverse=False)
```

Restrictions by purpose

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In [ ]:
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```

In [ ]:
```
sorted(iterable, /, *, key=None, reverse=False)
```

- arguments before `/` must be specified with position
- arguments after `*` must be specified with keyword

More on functions: Default and Optional arguments

How to define functions with default arguments

In [ ]:

```
def format_sentence(subject, value = "finished", end = "...."):
    return 'The ' + subject + ' is ' + value + end

format_sentence('lecture')
```

How to define functions with optional arguments

- Convention: use the object `None`

In [ ]:

```
def format_sentence(subject, value, end='.', second_value=None):
    if second_value is None:
        return 'The ' + subject + ' is ' + value + end
    else:
        return 'The ' + subject + ' is ' + value + ' and ' + second_valu

print(format_sentence('lecture', 'ongoing'))

#print(format_sentence('lecture', 'ongoing', second_value='self-referent
```

Summary

- `return` gives a value back, so you can repurpose it
    - a function can return several values at once: `return a, b`
    - a function without `return` gives back `None`
    -
- Types of arguments:
    - positional (**before** keyword)
    - keyword (**after** positional)
    - default (can be skipped)
    - optional (can be also skipped)

Exercise 1 (~30-40 minutes)

- Go to Canvas, `Modules -> Day 4 -> Exercise 1 - day 4`
- Extra reading:
    - https://realpython.com/python-kwargs-and-args/
    - https://able.bio/rhett/python-functions-and-best-practices--78aclaa

Documenting and commenting your code

It concerns two group of people:

- users: interested on how it is used
- developers: interested on how it works

Comments:

- mostly for developers. Describes the code
- pound sign (`#`) is used
- it is short
- tags can be used (BUG, TODO etc)

Documenting:

- mostly for users. Describes the use
- docstrings are used (`"""..."""`)
- it is quite long

Docstrings

In [ ]:

```
def process_file(filename, chrom, pos):
    """
    Read a very large vcf file, search for lines matching
    chromosome chrom and position pos.

    Print the genotypes of the matching lines.
    """
    for line in open(filename):
        if not line.startswith('#'):
            col = line.split('\t')
            if col[0] == chrom and int(col[1]) == pos:
                print(col[9:])
```

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            if col[0] == chrom and int(col[1]) == pos:
                print(col[9:])
```

In [ ]:
```
help(process_file)
```

Docstrings should always be in triple quotes even if they are short and one line

In [ ]:
```
import random
def make_list(x):
    # Returns a random list of length x.
    li = list(range(x))
    random.shuffle(li)
    return li
```

In [ ]:
```
help(make_list)
```

What to write in a docstring

- What the function does
- The parameters (and their default values)
- What it returns

In [ ]:

```python
def gc_content(sequence, decimals=2):
    """
    Calculate the GC content of a DNA sequence.

    Parameters:
        sequence: DNA sequence as a string, e.g. 'ATGC'
        decimals: number of decimals to round to (default 2)

    Returns:
        The fraction of G and C as a float, or None if the sequence is e
    """
    if len(sequence) == 0:
        return None
    gc = sequence.upper().count('G') + sequence.upper().count('C')
    return round(gc / len(sequence), decimals)
```

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```

In [ ]:
```python
help(gc_content)
```

Comments

Use comments wherever the code is hard to understand

In [ ]:
```
my_list[5] += other_list[3]  # explain why you do this!
```

Comments: explain *why*, not *what*

The code already says **what** happens. The comment should say **why**.

```
count += 1   # add 1 to count
```

Comments: explain *why*, not *what*

The code already says **what** happens. The comment should say **why**.

```
count += 1   # add 1 to count
```

```
rating = float(fields[1].strip())  # ratings in the file have leading spaces
```

A wrong comment is worse than no comment

```
# count the cytosines
count = sequence.count('G')
```

A wrong comment is worse than no comment

```
# count the cytosines
count = sequence.count('G')
```

- Which one is right, the comment or the code?
- When you change the code, update the comment too

Read more:

https://realpython.com/documenting-python-code/

comments: https://peps.python.org/pep-0008/#comments

docstrings: https://peps.python.org/pep-0257/

Error handling

Two paths for error handling:

1. Check conditions before performing a task

```
if x_can_be_done:
    do_this_with_x
else:
    error_handling
```

2. Perform the task and deal with errors afterwards

```
try:
    do_whatever_with_x
except <ErrorType>:
    error_handling
```

try and except: Check and report errors

Error handling

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1. Check conditions before performing a task

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if x_can_be_done:
    do_this_with_x
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2. Perform the task and deal with errors afterwards

```
try:
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except <ErrorType>:
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```

try and except: Check and report errors

Reason for catching errors: Check if I can recover from this error

Keep in mind: It is good to specify the error, e.g. `except ValueError:`

Try and except basic example

In [ ]:
```
int("five")
```

Try and except basic example

In [ ]:
```
int("five")
```

Your own message + Python's message

In [ ]:
```
try:
    int("five")
except ValueError:
    print("Error: try again")
```

Catching a Recoverable Error

Scenario: I am perfect

Input:

- My code is perfect
- The data I use is not perfect (e.g. some entries are invalid or missing)

Output:

- The code will throw an error and stop, making it impossible to process the remaining data

In [ ]:
```
with open('../files/bank_statement.txt', 'r', encoding = 'utf-8') as ban
    content = bank_file.read()
    print(content)
```

In [ ]:

```
fh = open("../files/bank_statement.txt", "r", encoding = "utf-8")

total = 0

for line in fh:
    expenses = line.strip().split()
    store = expenses[0]
    price = float(expenses[1])
    if store == 'ICA':
        total = total + price

fh.close()
print('Total amount spent on ICA is: '+str(total))
```

In [ ]:
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fh = open("../files/bank_statement.txt", "r", encoding = "utf-8")

total = 0

for line in fh:
    expenses = line.strip().split()
    store = expenses[0]
    price = float(expenses[1])
    if store == 'ICA':
        total = total + price

fh.close()
print('Total amount spent on ICA is: '+str(total))
```

Two main problems:

- The code is exiting but there is nothing wrong with it
- The file is not closed

In [ ]:
```
print(f"File closed after error? {fh.closed}")
```

Raise your own error

Scenario: Validating a variable

- At some point in my code, I want to stop execution if a variable is not an integer
- How can I interrupt the execution and raise an error?

In [ ]:

```
# Accept only integers
my_number=3.2

if type(my_number) is not int:
    print("Oooooohhhhh NOOOOOOOO")

print("more code follows")
```

User input

The code asks the user for input

- `input()`

In [ ]:
```
give_input = input("Give a number: ")

print(f'You pressed the number {give_input}')
```

Type of input

In [ ]:
```
number_in=input("Give me a number: ")

type(number_in)
```

Type of input

In [ ]:
```
number_in=input("Give me a number: ")

type(number_in)
```

The type of the input is **ALWAYS** string

Ensure correct type of input (exercise)

We want a script which asks the user to give an integer number. If the input of the user is not integer then inform about that and ask again about an integer number

- Think about it and we will do it together
- Hint: inputs have string type

In [ ]:

Summary

- Documentation
    - Docstrings (`"""..."""`) are for **users**: what the function does
    - Comments (`#`) are for **developers**: explain *why*, not *what*
- Error handling Comments (`#`) are for **developers**: explain *why*, not *what*
    - `try..except` catches an error, so your program does not crash
    - name the error you expect: `except ValueError:`
    - `except ValueError as err:` gives you Python's own error message
    - `raise` stops the program with your own error: `raise TypeError("The number is not integer")`
- `input()` asks the user for a value, and it **always** gives a string
    - convert it with `int()` or `float()`

Exercise 2 (~20-30 minutes)

- Go to Canvas, `Modules -> Day 4 -> Exercise 2 - day 4`

Lunch

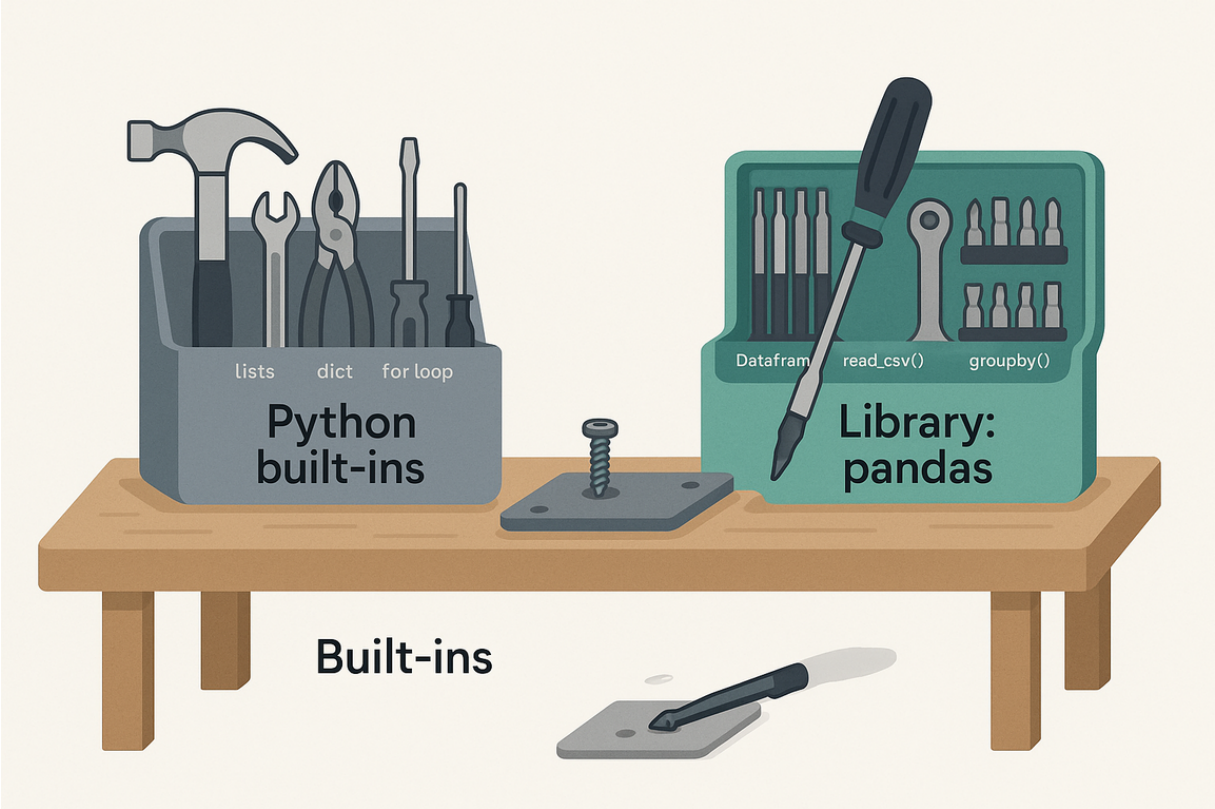
Pandas!!!



Pandas

- Library for working with tabular data
- Data analysis:
    - filter
    - transform
    - aggregate
    - plot
- Main hero: the `DataFrame` type

Built-in vs library tools - 1



Built-in vs library tools - 2

Task of the script:

- Load movies data from a file
- Calculate the average rating of all movies
- Filter the great movies: Rating >= 8.5
- Display them sorted by Rating (highest first), showing Title, Year, and Rating

Built-in solution

In [ ]:

```
# 1. LOAD DATA
movies = []
for line in open('../downloads/250.imdb', encoding='utf-8'):
    if line.startswith('#'):  # skip the header line
        continue
    fields = line.strip().split('|')
    rating = float(fields[1])
    year = int(fields[2])
    title = fields[6]
    movies.append([title, year, rating])

print(f"Loaded {len(movies)} movies")

# 2. AVERAGE RATING
total = 0
for movie in movies:
    total = total + movie[2]

average = total / len(movies)
print(f"\nAverage Rating: {average:.2f}")

# 3. FILTER Rating >= 8.5
top_movies = []
for movie in movies:
    if movie[2] >= 8.5:
        top_movies.append(movie)

# 4. SORT by Rating, highest first
def get_rating(movie):
    return movie[2]

top_movies.sort(key=get_rating, reverse=True)

print(f"\nGreat Movies (Rating >= 8.5): {len(top_movies)} found\n")
print(f"{'Title':<52} {'Year':>5} {'Rating':>7}")
print("-" * 66)
for movie in top_movies:
    print(f"{movie[0]:<52} {movie[1]:>5} {movie[2]:>7.1f}")
```

Library (pandas) solution

In [ ]:

```python
import pandas as pd

# 1. LOAD DATA
df = pd.read_csv('../downloads/250.imdb', sep='|')

# Strip spaces from ALL column names at once
df.columns = df.columns.str.strip()

# Rename '# Votes' to 'Votes'
df = df.rename(columns={'# Votes': 'Votes'})

print(f"Loaded {len(df)} movies")

# 2. AVERAGE RATING
print(f"\nAverage Rating: {df['Rating'].mean():.2f}")

# 3. & 4. FILTER + SORT
top_movies = (
    df[df['Rating'] >= 8.5]
    .sort_values('Rating', ascending=False)
)

print(f"\nGreat Movies (Rating >= 8.5): {len(top_movies)} found")
top_movies[['Title', 'Year', 'Rating']]
```

Built-in vs library tools - 3

Built-in Python — 27 lines

```python
# 1. LOAD DATA
movies = []
for line in open('../downloads/250.imdb', encoding='utf-8'):
    if line.startswith('#'):  # skip the header line
        continue
    fields = line.strip().split('|')
    rating = float(fields[1])
    year = int(fields[2])
    title = fields[6]
    movies.append([title, year, rating])

print(f"Loaded {len(movies)} movies")
```

```python
# 2. AVERAGE RATING
total = 0
for movie in movies:
    total = total + movie[2]

average = total / len(movies)
print(f"\nAverage Rating: {average:.2f}")
```

```python
# 3. FILTER Rating >= 8.5
top_movies = []
for movie in movies:
    if movie[2] >= 8.5:
        top_movies.append(movie)
```

```python
# 4. SORT by Rating, highest first
def get_rating(movie):
    return movie[2]

top_movies.sort(key=get_rating, reverse=True)
```

```python
print(f"\nGreat Movies (Rating >= 8.5): {len(top_movies)} found\n")
print(f"{'Title':<52} {'Year':>5} {'Rating':>7}")
print("-" * 66)
for movie in top_movies:
    print(f"{movie[0]:<52} {movie[1]:>5} {movie[2]:>7.1f}")
```

pandas — 12 lines

```python
import pandas as pd
```

```python
# 1. LOAD DATA
df = pd.read_csv('../downloads/250.imdb', sep='|')

# Strip spaces from ALL column names at once
df.columns = df.columns.str.strip()

# Rename '# Votes' to 'Votes'
df = df.rename(columns={'# Votes': 'Votes'})

print(f"Loaded {len(df)} movies")
```

```python
# 2. AVERAGE RATING
print(f"\nAverage Rating: {df['Rating'].mean():.2f}")
```

```python
# 3. & 4. FILTER + SORT
top_movies = (
    df[df['Rating'] >= 8.5]
    .sort_values('Rating', ascending=False)
)
```

```python
print(f"\nGreat Movies (Rating >= 8.5): {len(top_movies)} found")
top_movies[['Title', 'Year', 'Rating']]
```

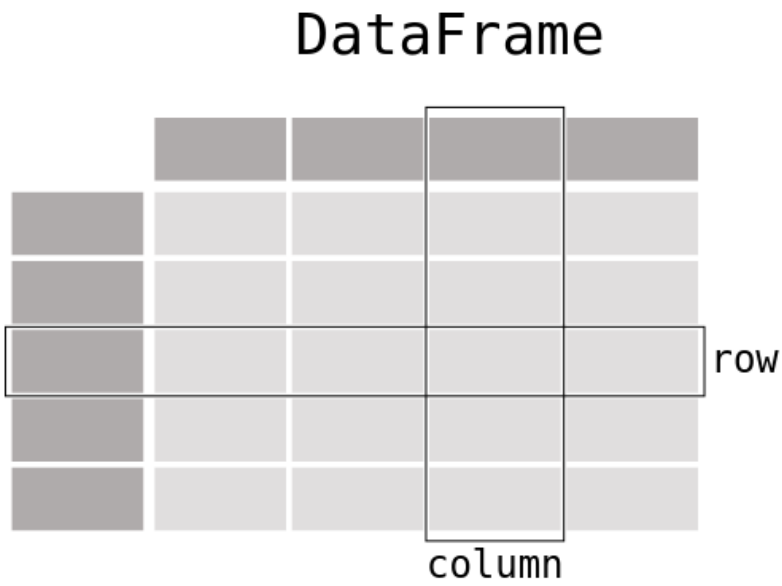
1. Load · 2. Average · 3. Filter · 4. Sort · Display (lines counted without comments and blank lines)

Benefits of shorter code

- Easier to read: you see **what** is done, not **how**
- Fewer lines, fewer places for bugs
- Easier to modify it: to sort by year, change `'Rating'` to `'Year'`

Pandas DataFrame

Pandas Dataframe is a **2D data structure**. In other words, is a table with rows and columns.



Creating a small DataFrame

In [ ]:
```
data = {
    'age': [1,2,3,4],
    'circumference': [2,3,5,10],
    'height': [30, 35, 40, 50]
}
```

In [ ]:
```
# Locate row
```

In [ ]:
```
# add name index
row_index = ["tree1", "tree2", "tree3", "tree4"]
```

In [ ]:
```
# Locate row
```

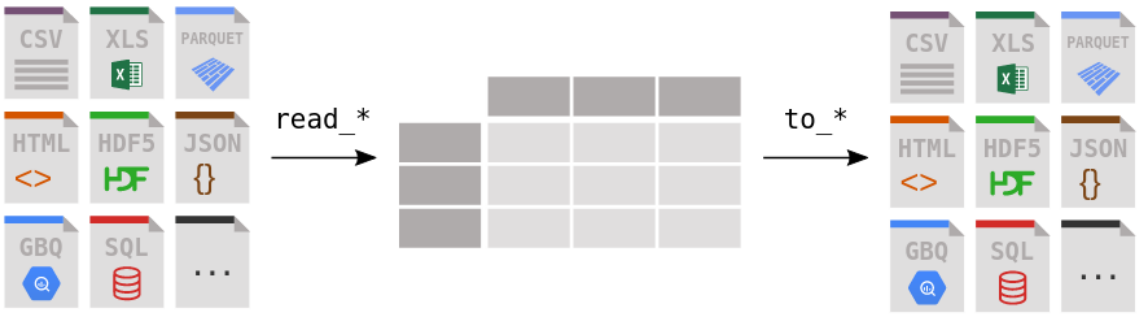
In [ ]:
```
# drop index
df = df.reset_index(drop=True)
df
```

Pandas can import data from many formats

- `pd.read_table` : tab separated values `.tsv`
- `pd.read_csv` : comma separated values `.csv`
- `pd.read_excel` : Excel spreadsheets `.xlsx`

DataFrame can export data to many formats

- `df.to_csv(sep='\t')` : tab separated values `.tsv`
- `df.to_csv()` : comma separated values `.csv`
- `df.to_excel()` : Excel spreadsheets `.xlsx`


CSV
XLS
PARQUET
HTML
HDF5
JSON
GBQ
SQL
...
read_*
to_*
CSV
XLS
PARQUET
HTML
HDF5
JSON
GBQ
SQL
...

Orange tree data

In [ ]:
```
df = pd.read_table('../downloads/Orange_1.tsv')
df
```

Read larger file

In [ ]:
```
df_big = pd.read_csv('../downloads/Orange_1_big.csv')
df_big
```

Read larger file

In [ ]:
```
df_big = pd.read_csv('../downloads/Orange_1_big.csv')
df_big
```

Do not panic

In [ ]:
```
pd.options.display.max_rows
```

Overview of your data, basic statistics

In [ ]:
```
df
```

Overview of your data, basic statistics

In [ ]:
```
df
```

In [ ]:
```
df.shape
```

Overview of your data, basic statistics

In [ ]:
```
df
```

In [ ]:
```
df.shape
```

In [ ]:
```
df.describe()
```

Overview of your data, basic statistics

In [ ]:
```
df
```

In [ ]:
```
df.shape
```

In [ ]:
```
df.describe()
```

In [ ]:
```
df.max()
```

Overview of your data, basic statistics

In [ ]:
```
df
```

In [ ]:
```
df.shape
```

In [ ]:
```
df.describe()
```

In [ ]:
```
df.max()
```

In [ ]:
```
df_big.head(10)
```

Overview of your data, basic statistics

In [ ]:
```
df
```

In [ ]:
```
df.shape
```

In [ ]:
```
df.describe()
```

In [ ]:
```
df.max()
```

In [ ]:
```
df_big.head(10)
```

In [ ]:
```
df_big.tail()
```

Overview of your data, basic statistics

In [ ]:
```
df
```

In [ ]:
```
df.shape
```

In [ ]:
```
df.describe()
```

In [ ]:
```
df.max()
```

In [ ]:
```
df_big.head(10)
```

In [ ]:
```
df_big.tail()
```

In [ ]:
```
df_big.info()
```

Selecting columns, rows and cells from a dataframe



Selecting one column

```
dataframe.columnname
dataframe['columnname']
```

Selecting one column

```
dataframe.columnname
dataframe['columnname']
```

In [ ]:
```
df
```

In [ ]:
```
df.age
```

In [ ]:
```
df['age']
```

Selecting one column

```
dataframe.columnname
dataframe['columnname']
```

In [ ]:
```
df
```

In [ ]:
```
df.age
```

In [ ]:
```
df['age']
```

`df.age` only works when the column name has no spaces and is not also the name of a method.

For example, a column called *count* returns the `.count()` method instead of the data.

So the **advice** is: `df['age']` always works, so prefer it.

Unique values of a column

In [ ]:
```
df['Tree'].unique()
```

Unique values of a column

In [ ]:
```
df['Tree'].unique()
```

How many of each?

In [ ]:
```
df['Tree'].value_counts()
```

Unique values of a column

In [ ]:
```
df['Tree'].unique()
```

How many of each?

In [ ]:
```
df['Tree'].value_counts()
```

- `unique()` gives the different values and `value_counts()` counts how often each one appears.
- Bioinformatics examples: variants per chromosome, reads per sample, genes per biotype.

Be careful with the headers of the column

In [ ]:
```
import pandas as pd
data = {
    'age ': [1,2,3,4],
    'circumference': [2,3,5,10],
    'height': [30, 35, 40, 50]
}
df_test = pd.DataFrame(data)
df_test
```

Be careful with the headers of the column

In [ ]:
```
import pandas as pd
data = {
    'age ': [1,2,3,4],
    'circumference': [2,3,5,10],
    'height': [30, 35, 40, 50]
}
df_test = pd.DataFrame(data)
df_test
```

In [ ]:
```
df_test.age
```

Be careful with the headers of the column

In [ ]:
```
import pandas as pd
data = {
    'age ': [1,2,3,4],
    'circumference': [2,3,5,10],
    'height': [30, 35, 40, 50]
}
df_test = pd.DataFrame(data)
df_test
```

In [ ]:
```
df_test.age
```

In [ ]:
```
df_test['age']
```

Strip the spaces

In [ ]:

Selecting multiple columns

In [ ]:
```
df
```

In [ ]:
```
df['age']
```

Selecting rows from a dataframe

In [ ]:
```
df
```

In [ ]:
```
df.loc[0] # select the first row
```

In [ ]:
```
df.loc[1:3] # select from row 2 to 4
```

In [ ]:
```
df.loc[1, 3] # select row 2 and 4
```

Selecting rows from a dataframe

In [ ]:
```
df
```

In [ ]:
```
df.loc[0] # select the first row
```

In [ ]:
```
df.loc[1:3] # select from row 2 to 4
```

In [ ]:
```
df.loc[1, 3] # select row 2 and 4
```

REMINDER:

- `.loc` uses the index labels, not the position of the row
- when range is used in `.loc` the *stop* part is included

Selecting cells from a dataframe

In [ ]:
```
df
```

In [ ]:
```
# Single cell
df.loc[0, 'age']

# One row, multiple columns

# Multiple rows, one column

# Multiple rows and columns
```

Run statistics on specific rows, columns, cells

In [ ]:
```
df[['age', 'circumference']].describe()
```

In [ ]:
```
df['age'].std()
```

In [ ]:
```
df.loc[0].describe()
```

In [ ]:
```
df.loc[0, 'age'] - df['age'].mean()
```

Selecting data from a dataframe by position

```
dataframe.iloc[row_position]
dataframe.iloc[row_position, column_position]
dataframe.iloc[start:stop]
```

Note: the *stop* part in the `.iloc` is not included (different from `.loc`)

Further reading from pandas documentation:
https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.iloc.html

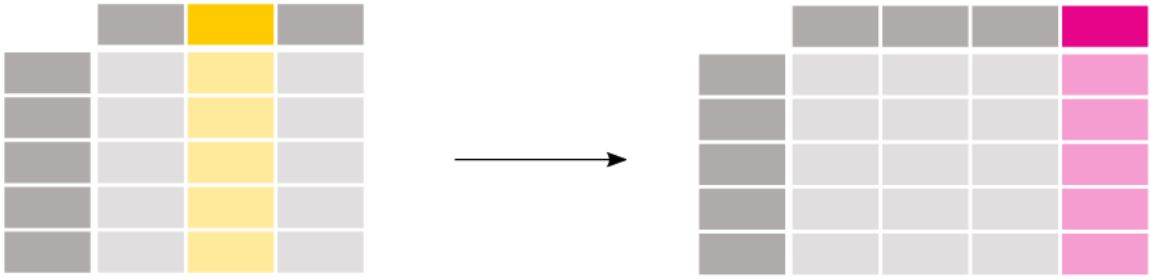
In [ ]:
```
df
```

In [ ]:
```
# Show the first column

# Show the second row

# Show the cell of the second row and the first column
```

Creating new column derived from existing column



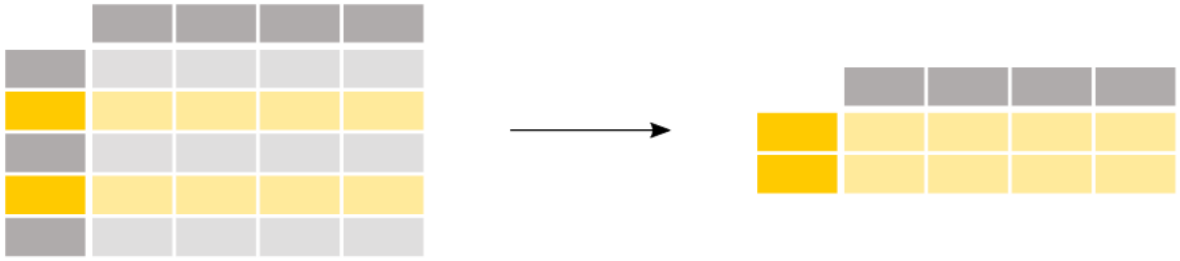
In [ ]:
```
import math
df['radius'] = df['circumference'] / (2.0 * math.pi)
df
```

Quiz time

Go to Canvas, `Modules -> Day 4 -> PyQuiz 4.1`

~5 min

Pandas concatenate



In [ ]:
```
df1 = pd.DataFrame({
    'age': [1,2,3,4],
    'circumference': [2,3,5,10],
    'height': [30, 35, 40, 50]
})

df1
```

In [ ]:
```
df2 = pd.DataFrame({
    'name': ['palm', 'ada', 'ek', 'olive'],
    'price': [1423, 2000, 102, 30]
})

df2
```

In [ ]:
```
df_con = pd.concat([df1,df2])
df_con
```

In [ ]:
```
# index is not reset and it will cause some problems
df_con.loc[0]
```

Reset index in concatenated dataframe

In [ ]:
```
df_con=df_con.reset_index(drop=True)
df_con
```

Control the concatenation axis:

- axis=0 (default): rows concatenation
- axis=1: columns concatenation

In [ ]:
```
pd.concat([df2, df1], axis=0).reset_index(drop=True)
```

In [ ]:
```
pd.concat([df2, df1], axis=1)
```

Selecting/filtering the dataframe by condition

e.g.

- Only trees with age larger than 100
- Only tree with circumference shorter than 20

In [ ]:
```
df = pd.read_table('../downloads/Orange.tsv')
df
```

Selecting with condition

`dataframe[`*`condition`*`]`

Selecting with condition

`dataframe[condition]`

In [ ]:
```
df[df['Tree'] == 1]
```

Selecting with condition

`dataframe[condition]`

In [ ]:
```
df[df['Tree'] == 1]
```

In [ ]:
```
df[df['age'] > 500]
```

Filter with multiple conditions

```
dataframe[(condition 1) & (condition 2)]
```

```
dataframe[(condition 1) | (condition 2)]
```

Filter with multiple conditions

`dataframe[(condition 1) & (condition 2)]`

`dataframe[(condition 1) | (condition 2)]`

In [ ]:
```
df[(df['age'] > 100) & (df['age'] <= 250)]
```

Filter with multiple conditions

`dataframe[(condition 1) & (condition 2)]`

`dataframe[(condition 1) | (condition 2)]`

In [ ]:
```
df[(df['age'] > 100) & (df['age'] <= 250)]
```

In [ ]:
```
df[(df['age'] > 500) & (df['circumference'] < 100) ]
```

Filter with multiple conditions

`dataframe[(condition 1) & (condition 2)]`

`dataframe[(condition 1) | (condition 2)]`

In [ ]:
```
df[(df['age'] > 100) & (df['age'] <= 250)]
```

In [ ]:
```
df[(df['age'] > 500) & (df['circumference'] < 100) ]
```

In [ ]:
```
df[(df['Tree'] == 1) | (df['Tree'] == 3)]
```

Filter with multiple conditions

`dataframe[(condition 1) & (condition 2)]`

`dataframe[(condition 1) | (condition 2)]`

In [ ]:
```
df[(df['age'] > 100) & (df['age'] <= 250)]
```

In [ ]:
```
df[(df['age'] > 500) & (df['circumference'] < 100) ]
```

In [ ]:
```
df[(df['Tree'] == 1) | (df['Tree'] == 3)]
```

- `&` means "and": both conditions must be true
- `|` means "or": at least one condition must be true
- Put each condition in brackets ( )
- Do not use the Python words **and**/**or** here

Small exercise 1

- Find the maximal circumference and then filter the dataframe by it

In [ ]:
```
df
```

Statistics per group

Question: what is the maximum circumference of each tree?

In [ ]:

Statistics per group

Question: what is the maximum circumference of each tree?

In [ ]:

- `groupby('Tree')`: split the table into one group per tree
- `['circumference']`: choose the column
- `.max()`: calculate the maximum for each group; the results are combined into one table

Bioinformatics examples: the mean expression per gene, the number of variants per sample.

Saving your results to a file

In [ ]:
```
# create table
old_trees = df[df['age'] > 1000]
```

In [ ]:
```
# read table
```

Quiz time

Go to Canvas, `Modules -> Day 4 -> PyQuiz 4.2`

Plotting

- pandas can plot your data directly
- One pattern for all plots:

```
df.plot(x='column', y='column', kind='...')
```

- More information:
  https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.plot.html

Which plot for which question?

`kind`	Use it for	Bioinformatics example
`line`	change over time or in order	growth, time course
`scatter`	relationship between two numbers	expression of gene A vs gene B
`hist`	distribution of one variable	read lengths, quality scores
`box`	comparing distributions between groups	expression per condition
`bar`	comparing categories	number of variants per chromosome

The data we will plot

In [ ]:
```
import pandas as pd

df = pd.read_table('../downloads/Orange.tsv')
df_big = pd.read_csv('../downloads/Orange_1_big.csv')
```

Line plot

In [ ]:

```
tree1 = df[df['Tree'] == 1]
tree1
# Simplest plot: df.columnname.plot() - default=line


# Nicer plot: set x and y axis, kind, title, labels
```

Scatter plot

In [ ]:
```
df.plot(kind='scatter', x='age', y='circumference',
        title='Age vs circumference',
        xlabel='Age (days)', ylabel='Circumference (mm)')
```

Histogram

In [ ]:
```
df_big.plot(kind='hist', y='height', bins=15,
            title='Distribution of tree height', xlabel='Height')
```

Box plot

In [ ]:
```
df.boxplot(column='circumference', by='Tree')
```

Bar plot

In [ ]:
```
df.groupby('Tree')['circumference'].max().plot(kind='bar',
        title='Maximum circumference per tree',
        xlabel='Tree', ylabel='Circumference (mm)')
```

Size and font

In [ ]:
```
# figsize and fontsize
df.plot(kind='scatter', x='age', y='circumference',
        title='Age vs circumference',
        xlabel='Age (days)', ylabel='Circumference (mm)',
        fontsize=16, figsize=(12, 8))
```

Build your plot step by step

In [ ]:
```
df.plot(kind='scatter', x='age', y='circumference',
        figsize=(12, 8), fontsize=16)
```

Saving a plot

In [ ]:
```
ax = df.plot(kind='scatter', x='age', y='circumference',
             title='Age vs circumference',
             xlabel='Age (days)', ylabel='Circumference (mm)')

ax.figure.savefig('orange_scatter.png', dpi=300)
```

Exercise 3 (~30 minutes)

- Go to Canvas, `Modules -> Day 4 -> Exercise 3 - day 4`

INFO: Re-download `downloads.zip` from the pre-course material if you do not have the `hypoxia_rnaseq.xlsx` in the downloads folder